

Lecture 5: 9/8/2026

Lecturer: Karthik Gajulapalli

Scribe: Wil Nguyen

Last Class

A proposition is a statement that is either true or false.

2 is even.

A predicate takes in input(s) and outputs True or False.

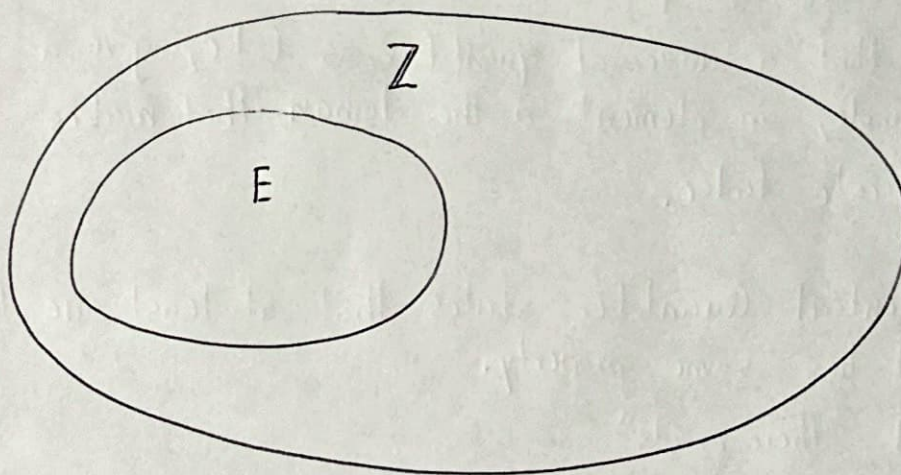
Predicates describe properties of objects.'

Predicates can be used to construct subsets.

Let $\cancel{N} = \{0, 1, 2, \dots\}$

$\mathbb{Z}$ be the set of integers

E be the set of even numbers



Quantifiers allow us to make propositions about many objects at once.'

Universal quantifier makes a statement about all elements in a set.

Symbol: $\forall$ "for all"

For all elements in the set of even numbers, those elements are in the set of integers.

$$(\forall x \in E) x \in \mathbb{Z} \quad (\in \text{ means in})$$

This is the same as:

$$x_1 \in \mathbb{Z} \wedge x_2 \in \mathbb{Z} \wedge x_3 \in \mathbb{Z} \wedge x_4 \in \mathbb{Z} \wedge \dots$$

where $x_1, x_2, x_3, x_4, \dots$ are in E

This can be written in as:

$$\bigwedge_{x \in E} x \in \mathbb{Z}$$

To prove that ~~an~~ a universal quantifier is true, choose an arbitrary, not special, element ~~of~~ in the domain, and show that the predicate is true.

To prove that a universal quantifier is false, give a counterexample, an element in the domain that makes the predicate false.

The Existential Quantifier states that at least one element in a set has some property.

Symbol: $\exists$ "there exists"

$$\exists x \in \mathbb{Z} \mid x \in E \equiv x_1 \in E \vee x_2 \in E \vee x_3 \in E \vee x_4 \in E \vee \dots$$

such
that

logical equivalence
(same truth value)

$$\equiv \bigvee_{x \in \mathbb{Z}} x \in E$$

To prove true, give witness, one element in domain with property.

To prove false, choose arbitrary element in domain and show that it cannot have property.

Since $\forall$ can be represented by AND's and $\exists$ can be represented by OR's, we can use De Morgan's law to switch between the two.

Let $P(x)$ be a predicate

D be a set

$x_1, x_2, x_3, \dots \in D$

$$(\forall x \in D) P(x) \equiv P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots$$

$$\equiv \neg [\neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots] \quad \text{De Morgan's}$$

$$\equiv \neg (\exists x \in D) \neg P(x)$$

This Class

~~Set builder notation~~

A set is a collection of objects.

The objects in a set are called elements.

An element can only appear once in a set.

The order of elements in a set does not matter.

~~$\{1, 1\}$~~

$\in$ means in $1 \in \{1, 2\}$

$\notin$ means not in $3 \notin \{1, 2\}$

sets can contain other sets

$\{\{1, 2\}, \{2\}, 1\}$

$\emptyset = \{\}$ (empty set)

Using set builder notation to construct a set:

$S = \{x \in \mathbb{N} \mid 10 \leq x \leq 100\} = \overset{\text{set of}}{\text{natural}} \text{ between } 10 \text{ and } 100$
element domain such that inclusive

Cardinality is the number of elements in a set.

$$|\emptyset| = 0$$

$$|\{1, 2\}| = 2$$

If there is a set in a set, the entire inner set is counted as 1 element.

$$|\{\{1, 2\}\}| = 1$$

$$|\{\{2, 3, 5\}, \{7\}, 5\}| = 3$$

Subset

$$\text{Let } B = \{1, 2, 3\}$$

Set A is a subset of set B means all elements in A are also in B .

$A \subseteq B$ means A is a subset of B .

$A \not\subseteq B$ means A is not a subset of B .

$$\text{If } A = \{1\} \quad A \subseteq B$$

$$C = \{2, 5\} \quad C \not\subseteq B$$

$$A \subseteq B \text{ if and only if } (\forall x \in A) x \in B$$

$\emptyset \subseteq B$ Since the empty set has no elements, there is no possible counterexample to show $\emptyset \not\subseteq B$.

Therefore, $\emptyset \subseteq B$ is vacuously true.

A is a proper subset of B if there A is a subset of B and B has more elements than A .

$\subset$ means proper subset

$$A \subset B \text{ if and only if } [(\forall x \in A)(x \in B) \wedge (\exists y \in B | y \notin A)]$$

$$B = \{1, 2, 3\}$$

$$A = \{1, 2, 3\}$$

$$A \subseteq B$$

$$A \not\subseteq B$$

A is subset of B

A is not proper subset of B

$$A = B \Leftrightarrow A \subseteq B \wedge B \subseteq A$$

A does not have any element not in B

A does not have any more

A is not missing any element in B

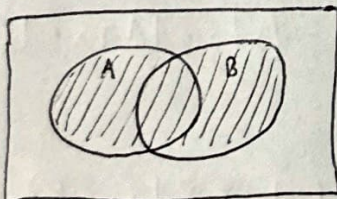
A does not have any less

A and B must be the same

Operations on sets

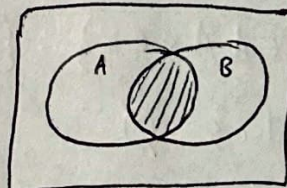
A union B

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



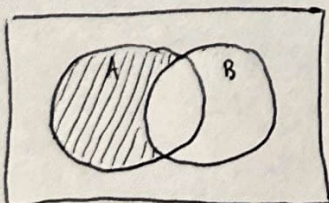
A intersect B

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



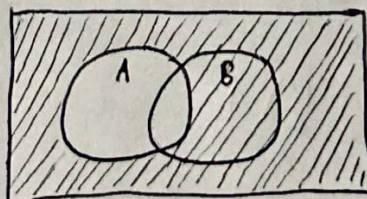
A difference B

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}$$



A complement

$$\bar{A} = \{x \mid x \notin A\}$$



$$A \cup B = \{x \mid x \in A \vee x \in B\} = \{x \mid x \in B \vee x \in A\} = B \cup A$$

$$A \cap B = \{x \mid x \in A \wedge x \in B\} = \{x \mid x \in B \wedge x \in A\} = B \cap A$$

$$\cancel{A \cup B = \{x}}$$

$$A \cup \emptyset = \{x \mid x \in A \vee x \in \emptyset\} = \{x \mid x \in A \vee \text{False}\} = \{x \mid x \in A\} = A$$

$$A \cap \emptyset = \{x \mid x \in A \wedge x \in \emptyset\} = \{x \mid x \in A \wedge \text{False}\} = \{x \mid \text{False}\} = \emptyset$$

$$A \setminus \emptyset = \{x \mid x \in A \wedge x \notin \emptyset\} = \{x \mid x \in A \wedge \text{True}\} = \{x \mid x \in A\} = A$$

$$A \cup (B \cap C) = \{x \mid x \in A \vee x \in (B \cap C)\}$$

$$= \{x \mid x \in A \vee x \in \{x \mid x \in B \wedge x \in C\}\}$$

$$= \{x \mid x \in A \vee (x \in B \wedge x \in C)\}$$

$$= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} \text{ distributive}$$

$$= \{x \mid (x \in \{x \mid x \in A \vee x \in B\}) \wedge (x \in \{x \mid x \in A \vee x \in C\})\}$$

$$= \{x \mid (x \in (A \cup B)) \wedge (x \in (A \cup C))\}$$

$$= (A \cup B) \cap (A \cup C)$$

$$\text{If } A = B$$

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\} = \{x \mid x \in A \wedge x \notin A\} = \{x \mid x \in A \wedge \neg x \in A\}$$

$$= \{x \mid \text{False}\} = \emptyset$$

$$B \setminus A = \{x \mid x \in B \wedge x \notin A\} = \{x \mid x \in B \wedge x \notin B\} = \{x \mid x \in B \wedge \neg x \in B\}$$

$$= \{x \mid \text{False}\} = \emptyset$$

$$A \setminus B = B \setminus A$$

$$A = B \Rightarrow A \setminus B = B \setminus A$$

Inclusion and Exclusion

Let:

G = set of people all people in coding club

M = set of all people in music club

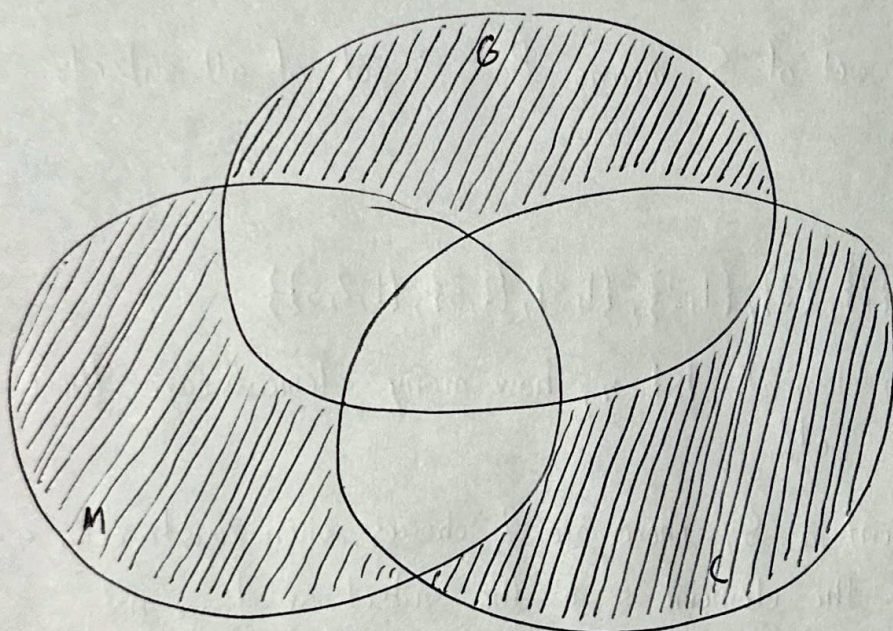
C = set of all people in chess club

$$|G| = 60 \quad |G \cap M| = 30 \quad |G \cap M \cap C| = 10$$

$$|M| = 50 \quad |G \cap C| = 25$$

$$|C| = 40 \quad |M \cap C| = 20$$

How many people are in exactly 1 club?



$$|G \setminus M \setminus C| + |M \setminus G \setminus C| + |C \setminus G \setminus M| = ?$$

$$|G \setminus M \setminus C| = |G| - |G \cap C| - |G \cap M| + |G \cap M \cap C|$$

number of people
in only coding

number of people
in coding and
chess

number of
people in coding
and music

number of people in
coding, music, and chess

we don't want to count these
people, they were counted in G so
we are going to subtract them

counted once in G , then
subtracted twice, so add
once to compensate

$$|G \setminus M \setminus C| = 60 - 25 - 30 + 10 = 15$$

$$|M \setminus G \setminus C| = |M| - |G \cap M| - |M \cap C| + |G \cap M \cap C|$$

$$= 50 - 30 - 20 + 10 = 10$$

$$|C \setminus G \setminus M| = |C| - |G \cap C| - |M \cap C| + |G \cap M \cap C|$$

$$= 40 - 25 - 20 + 10 = 5$$

$$15 + 10 + 5 = 30$$

There are 30 people in exactly 1 club.

Power set

P is a power set of S means P is a set of all subsets of S .

$$S = \{1, 2, 3\}$$

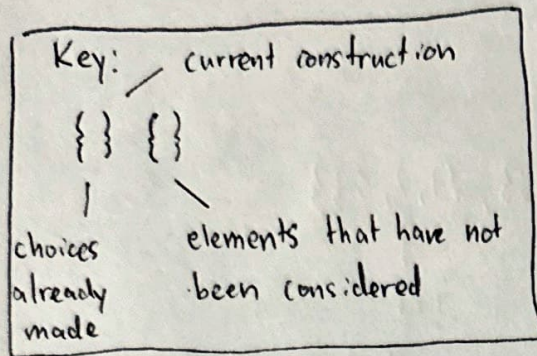
$$P = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

If S has n elements, $|S| = n$, how many elements are there in the power set?

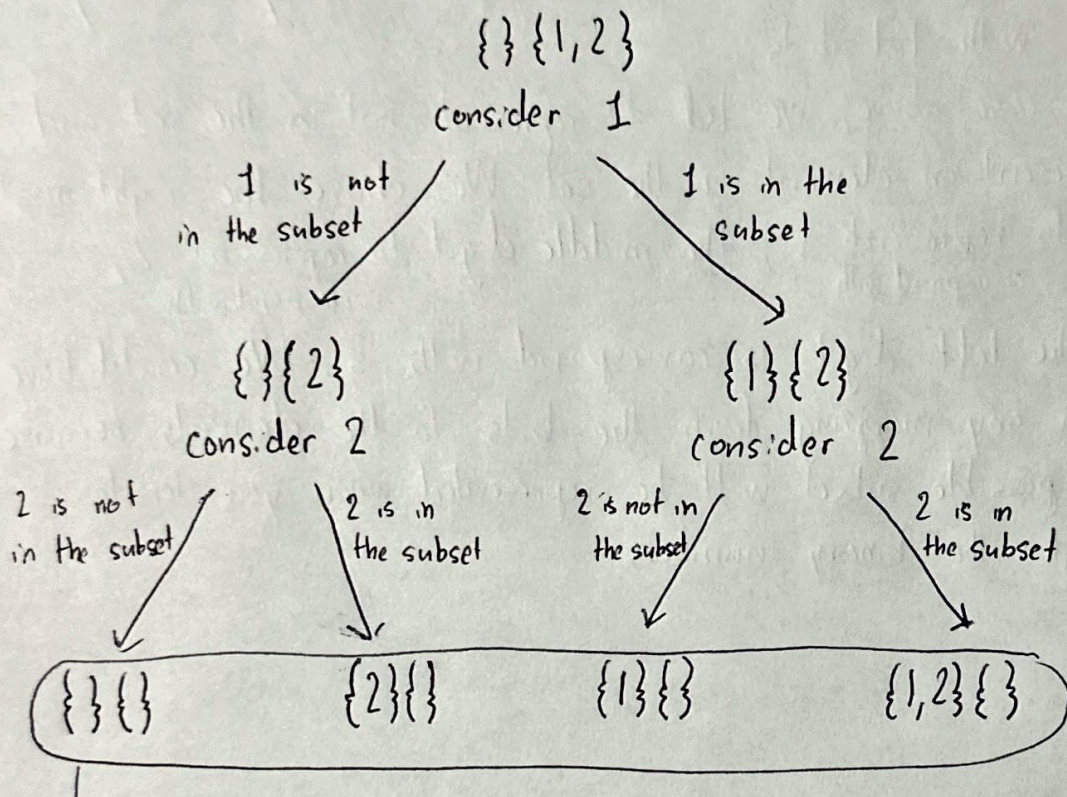
For each element in S , there are 2 choices when constructing a subset, either the element is in the subset or it is not.

Therefore, there are 2^n possible subsets.

$$|P| = 2^{|S|}$$



This is a recursive algorithm.
 The problem is broken down into a smaller problem until you get all possible subsets.
 Below is for power set of $\{1, 2\}$



These are all the possible subsets.

These constructions are complete, no more options to consider.

Click [here](#) to view code for recursive algorithm in python

Bit mapping

$$S = \{1, 2, 3\}$$

$$P = \{\{\}, \{1\}, \{2\}, \{1, 2\}, \{3\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

/				\			
000	001	010	011	100	101	110	111

Every possible subset of S can be represented by a binary number with 3 bits.

In the case above, we let 0 represent not in the set and 1 represent an element in the set. We chose the right most digit to represent 1, the middle digit to represent 2, and the left digit to correspond with 3.

We could have chosen any mapping from the bits to the elements because every possible subset will be represented when you iterate through the binary numbers.

[Click here to view bit mapping algorithm code in python](#)

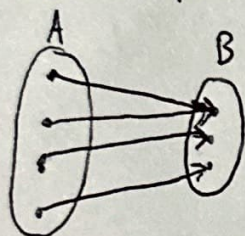
Are some infinities bigger than others?

We need a way to compare the number of elements in two sets.

If ~~all~~ Consider $A \rightarrow B$, A and B are sets

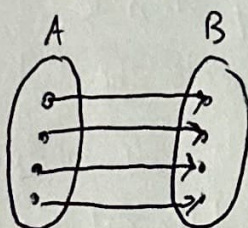
If ^{mapping} (all ^{elements in output space} outputs are mapped to): $|A| \geq |B|$

Case 1: $|A| > |B|$



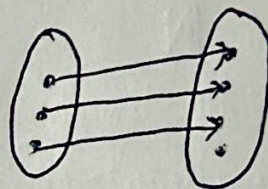
mapping
possible

Case 2: $|A| = |B|$



mapping
possible

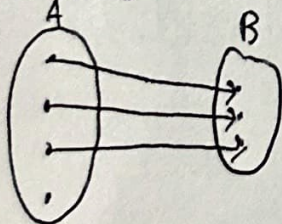
Case 3: $|A| < |B|$



mapping not possible
because ^{each} input can only
have 1 output

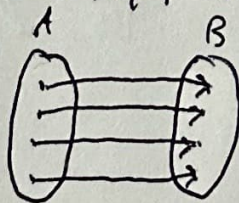
If (different input, then different output);

Case 1: $|A| > |B|$



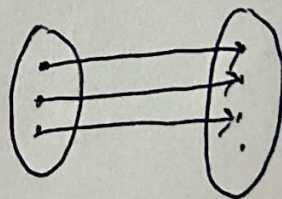
mapping not possible,
leftover input needs different
output but no different
outputs available

Case 2: $|A| = |B|$



mapping
possible

Case 3: $|A| < |B|$



mapping
possible

If (all outputs elements of output space mapped to
AND

different inputs give different outputs)

THEN $|A|=|B|$ (same number of elements)

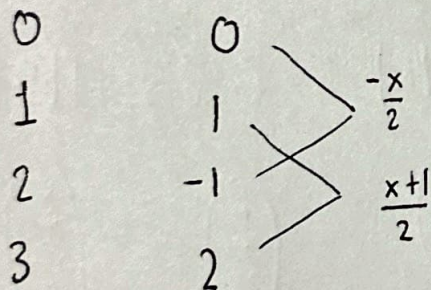
Countably infinite

A set S is countably infinite if there exists a mapping from $\mathbb{N}$ to S that meets the following conditions:

- All elements of output space mapped to
- Different inputs have different outputs

Is $\mathbb{Z}$ countably infinite?

$$\mathbb{N} \quad \mathbb{Z} \quad f(x) = \begin{cases} \frac{-x}{2} & \text{where } x \text{ is even} \\ \frac{x+1}{2} & \text{where } x \text{ is odd} \end{cases}$$



Yes, $\mathbb{Z}$ is countably infinite.

Are the real numbers countably infinite?

Assume $S = \{x \in \mathbb{R} \mid 0 < x < 1\}$ is countably infinite.

Then, ~~we can make~~ there exists a mapping from the natural numbers to S such that all elements in S are mapped to.

0	0.	3	4	1	6	0	...
1	0.	4	8	9	2	9	...
2	0.	4	2	4	5	6	...
3	0.	8	1	9	7	2	...
4	0.	1	4	3	7	8	...
⋮							

Construct a number such that the number has at least 1 digit different than all others from each real number in list.

0.56721
 $\begin{array}{ccccc} \swarrow & | & | & | & \searrow \\ 5 \neq 3 & 6 \neq 8 & 7 \neq 2 & 2 \neq 7 & 1 \neq 8 \end{array}$
 $7 \neq 4$

Make sure that new number does not end with repeating 0's or 9's to avoid for example $0.2 = 0.1\bar{9}$

New number is in S but not in list.

Contradiction, all elements in S should be in list.

S is not countably infinite.

$S \subset \mathbb{R}$ (the set of real numbers is at least as big as S)

$\mathbb{R}$ is not countably infinite.

Are there any restrictions on how you can define a set?

Russel's Paradox

Let $S = \{x \mid x \notin x\}$

S is the set of all sets that do not contain themselves.

Does S contain itself?

If it does, it must satisfy the condition, $S \notin S$.

If it does it doesn't. Contradiction

If it doesn't, it satisfies the condition to be in S .

If it doesn't it does. ~~Contradiction~~

Since we get a contradiction whether or not S contains itself, we cannot have no restrictions on defining sets.

References

[1] ~~Stand~~ Stanford CS103, Lecture 3 slides

Questions

- 1) How can we prove that the bit mapping algorithm can represent all possible subsets?
- 2) How can we prove the following:
$$(A \setminus B = B \setminus A) \Rightarrow A = B$$
- 3) If there are two different types of stamps, one ~~with~~ ^{worth} n cents and one ~~with~~ ^{worth} m cents, what is the minimum value x cents such that any integer cent value of at least x cents can be represented by the stamps?
- 4) Can you use induction to reverse engineer a solution?
- 5) ~~Show why~~ ^{Why does} rewriting $ax^2 + bx + c = 0$ as $ax^2 + dxe + c = 0$ where $a, b, c, d, e \neq 0$ and $dte = b$ and $d \cdot e = a \cdot c$ helps you factor?

COT 3100 — Lecture 5

Typed Version of the Handwritten Notes

Lecture 5: 9/8/2026
Lecturer: Karthik Gajulapalli
Scribe: Wil Nguyen

If there are any inaccuracies or if you have any questions, please email me at wi500104@ucf.edu

Source of this typed version. This typed section was produced by transcribing the handwritten Lecture 5 notes that appear on the preceding pages of this document. All text, equations, diagrams, and examples below are typed reproductions of those handwritten notes, scribed by Wil Nguyen for the lecture given by Karthik Gajulapalli on 9/8/2026. See reference [2].

Last Class

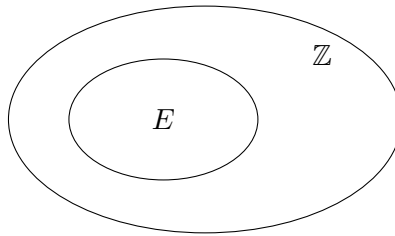
A **proposition** is a statement that is either true or false.

2 is even.

A **predicate** takes in input(s) and outputs True or False. Predicates describe properties of objects. Predicates can be used to construct subsets.

Let

- $\mathbb{Z}$ be the set of integers,
- E be the set of even numbers.



Quantifiers allow us to make propositions about many objects at once.

Universal Quantifier

The universal quantifier makes a statement about all elements in a set.

Symbol: $\forall$ “for all”

For all elements in the set of even numbers, those elements are in the set of integers.

$$(\forall x \in E) x \in \mathbb{Z} \quad (\in \text{ means in})$$

This is the same as:

$$x_1 \in \mathbb{Z} \wedge x_2 \in \mathbb{Z} \wedge x_3 \in \mathbb{Z} \wedge x_4 \in \mathbb{Z} \wedge \cdots$$

where $x_1, x_2, x_3, x_4, \dots$ are in E .

This can be written as:

$$\bigwedge_{x \in E} x \in \mathbb{Z}$$

To prove that a universal quantifier is **true**, choose an arbitrary, not special, element in the domain, and show that the predicate is true.

To prove that a universal quantifier is **false**, give a counterexample, an element in the domain that makes the predicate false.

Existential Quantifier

The Existential Quantifier states that at least one element in a set has some property.

Symbol: $\exists$ “there exists”

$$\exists x \in \mathbb{Z} \mid x \in E \equiv x_1 \in E \vee x_2 \in E \vee x_3 \in E \vee x_4 \in E \vee \dots$$

where $\mid$ means “such that” and $\equiv$ denotes logical equivalence (same truth value).

$$\equiv \bigvee_{x \in \mathbb{Z}} x \in E$$

To prove **true**, give a witness, one element in the domain with the property.

To prove **false**, choose an arbitrary element in the domain and show that it cannot have the property.

De Morgan’s Law for Quantifiers

Since $\forall$ can be represented by AND’s and $\exists$ can be represented by OR’s, we can use De Morgan’s law to switch between the two.

Let $P(x)$ be a predicate, D be a set, and $x_1, x_2, x_3, \dots \in D$.

$$\begin{aligned} (\forall x \in D) P(x) &\equiv P(x_1) \wedge P(x_2) \wedge P(x_3) \wedge \dots \\ &\equiv \neg [\neg P(x_1) \vee \neg P(x_2) \vee \neg P(x_3) \vee \dots] && \text{De Morgan’s} \\ &\equiv \neg (\exists x \in D) \neg P(x) \end{aligned}$$

This Class

Sets

A **set** is a collection of objects. The objects in a set are called **elements**.

An element can only appear once in a set. $\{1, 1\}$ **✗** (not allowed)

The order of elements in a set does not matter.

- $\in$ means *in* $1 \in \{1, 2\}$
- $\notin$ means *not in* $3 \notin \{1, 2\}$

Sets can contain other sets:

$$\{\{1, 2\}, \{2\}, 1\}$$

$$\emptyset = \{\}$$
 (empty set)

Set Builder Notation

Using set builder notation to construct a set:

$$S = \{ \underbrace{x}_{\text{element}} \in \underbrace{\mathbb{N}}_{\text{domain}} \mid \underbrace{10 \leq x \leq 100}_{\text{such that}} \} = \text{set of natural between 10 and 100 inclusive}$$

Cardinality

Cardinality is the number of elements in a set.

$$|\emptyset| = 0 \quad |\{1, 2\}| = 2$$

If there is a set in a set, the entire inner set is counted as 1 element.

$$|\{\{1, 2\}\}| = 1 \quad |\{\{2, 3, 5\}, \{7\}, 5\}| = 3$$

Subset

Let $B = \{1, 2, 3\}$.

Set A is a **subset** of set B means all elements in A are also in B .

- $A \subseteq B$ means A is a subset of B .
- $A \not\subseteq B$ means A is not a subset of B .

$$\begin{array}{ll} A = \{1\} & A \subseteq B \\ C = \{2, 5\} & C \not\subseteq B \end{array}$$

$$A \subseteq B \text{ if and only if } (\forall x \in A) x \in B$$

$\emptyset \subseteq B$. Since the empty set has no elements, there is no possible counterexample to show $\emptyset \not\subseteq B$. Therefore, $\emptyset \subseteq B$ is **vacuously true**.

A is a **proper subset** of B if A is a subset of B and B has more elements than A .

$\subset$ means proper subset.

$$A \subset B \text{ if and only if } [(\forall x \in A)(x \in B) \wedge (\exists y \in B \mid y \notin A)]$$

$$B = \{1, 2, 3\} \quad A = \{1, 2, 3\}$$

$$\underbrace{A \subseteq B}_{\substack{A \text{ is subset} \\ \text{of } B}}$$

$$\underbrace{A \not\subseteq B}_{\substack{A \text{ is not proper} \\ \text{subset of } B}}$$

Set Equality

$$A = B \iff A \subseteq B \wedge B \subseteq A$$

A does not have any
element not in B



A does not
have any more

A is not missing
any element in B



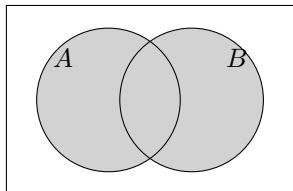
A does not
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A and B must be the same

Operations on Sets

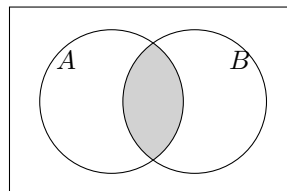
A union B

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$



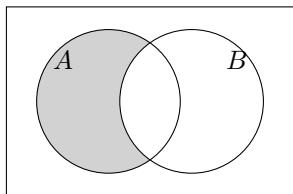
A intersect B

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$



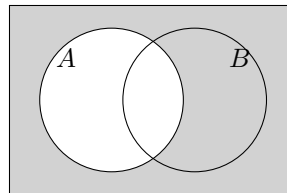
A difference B

$$A \setminus B = \{x \mid x \in A \wedge x \notin B\}$$



A complement

$$\overline{A} = \{x \mid x \notin A\}$$



Identities

$$A \cup B = \{x \mid x \in A \vee x \in B\} = \{x \mid x \in B \vee x \in A\} = B \cup A$$

$$A \cap B = \{x \mid x \in A \wedge x \in B\} = \{x \mid x \in B \wedge x \in A\} = B \cap A$$

$$\begin{aligned}
A \cup \emptyset &= \{x \mid x \in A \vee x \in \emptyset\} = \{x \mid x \in A \vee \text{False}\} = \{x \mid x \in A\} = A \\
A \cap \emptyset &= \{x \mid x \in A \wedge x \in \emptyset\} = \{x \mid x \in A \wedge \text{False}\} = \{x \mid \text{False}\} = \emptyset \\
A \setminus \emptyset &= \{x \mid x \in A \wedge x \notin \emptyset\} = \{x \mid x \in A \wedge \text{True}\} = \{x \mid x \in A\} = A
\end{aligned}$$

$$\begin{aligned}
A \cup (B \cap C) &= \{x \mid x \in A \vee x \in (B \cap C)\} \\
&= \{x \mid x \in A \vee x \in \{x \mid x \in B \wedge x \in C\}\} \\
&= \{x \mid x \in A \vee (x \in B \wedge x \in C)\} \\
&= \{x \mid (x \in A \vee x \in B) \wedge (x \in A \vee x \in C)\} \quad \text{distributive} \\
&= \{x \mid (x \in \{x \mid x \in A \vee x \in B\}) \wedge (x \in \{x \mid x \in A \vee x \in C\})\} \\
&= \{x \mid (x \in (A \cup B)) \wedge (x \in (A \cup C))\} \\
&= (A \cup B) \cap (A \cup C)
\end{aligned}$$

If $A = B$:

$$\begin{aligned}
A \setminus B &= \{x \mid x \in A \wedge x \notin B\} = \{x \mid x \in A \wedge x \notin A\} = \{x \mid x \in A \wedge \neg x \in A\} \\
&= \{x \mid \text{False}\} = \emptyset \\
B \setminus A &= \{x \mid x \in B \wedge x \notin A\} = \{x \mid x \in B \wedge x \notin B\} = \{x \mid x \in B \wedge \neg x \in B\} \\
&= \{x \mid \text{False}\} = \emptyset
\end{aligned}$$

$$\begin{aligned}
A \setminus B &= B \setminus A \\
A = B &\implies A \setminus B = B \setminus A
\end{aligned}$$

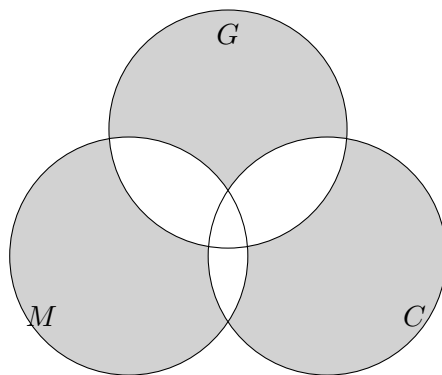
Inclusion and Exclusion

Let:

- G = set of all people in coding club
- M = set of all people in music club
- C = set of all people in chess club

$$\begin{aligned}
|G| &= 60 & |G \cap M| &= 30 & |G \cap M \cap C| &= 10 \\
|M| &= 50 & |G \cap C| &= 25 \\
|C| &= 40 & |M \cap C| &= 20
\end{aligned}$$

How many people are in exactly 1 club?



$$|G \setminus M \setminus C| + |M \setminus G \setminus C| + |C \setminus G \setminus M| = ?$$

$$\underbrace{|G \setminus M \setminus C|}_{\text{number of people in only coding}} = |G| - \underbrace{|G \cap C|}_{\text{number of people in coding and chess}} - \underbrace{|G \cap M|}_{\text{number of people in coding and music}} + \underbrace{|G \cap M \cap C|}_{\text{number of people in coding, music, and chess}}$$

$|G \cap C|$ and $|G \cap M|$: we don't want to count these people, they were counted in G so we are going to subtract them.

$|G \cap M \cap C|$: counted once in G , then subtracted twice, so add once to compensate.

$$|G \setminus M \setminus C| = 60 - 25 - 30 + 10 = 15$$

$$\begin{aligned} |M \setminus G \setminus C| &= |M| - |G \cap M| - |M \cap C| + |G \cap M \cap C| \\ &= 50 - 30 - 20 + 10 = 10 \end{aligned}$$

$$\begin{aligned} |C \setminus G \setminus M| &= |C| - |G \cap C| - |M \cap C| + |G \cap M \cap C| \\ &= 40 - 25 - 20 + 10 = 5 \end{aligned}$$

$$15 + 10 + 5 = 30$$

There are 30 people in exactly 1 club.

Power Set

P is a **power set** of S means P is a set of all subsets of S .

$$S = \{1, 2, 3\}$$

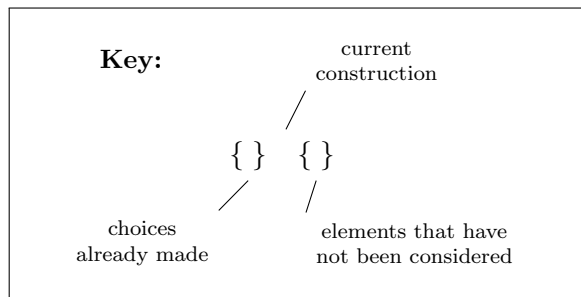
$$P = \{ \{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \}$$

If S has n elements, $|S| = n$, how many elements are there in the power set?

For each element in S , there are 2 choices when constructing a subset, either the element is in the subset or it is not. Therefore, there are 2^n possible subsets.

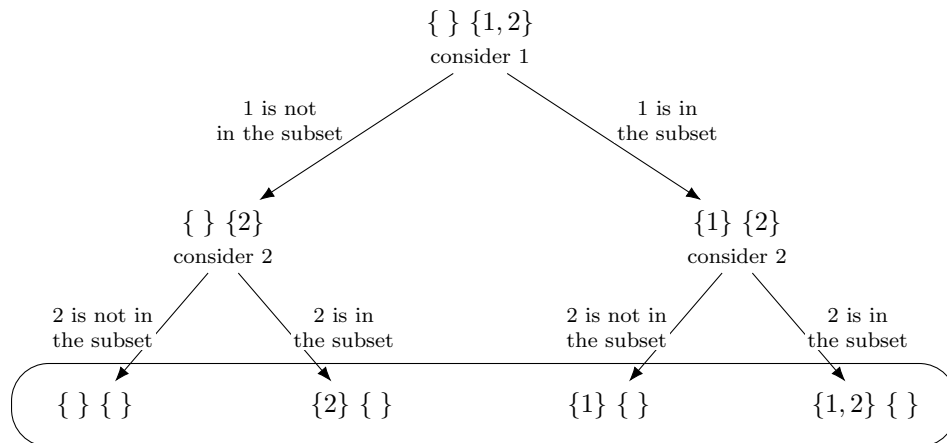
$$|P| = 2^{|S|}$$

Recursive Algorithm



This is a recursive algorithm. The problem is broken down into a smaller problem until you get all possible subsets.

Below is for the power set of $\{1, 2\}$.



These are all the possible subsets. These constructions are complete, no more options to consider.

[Click here to view code for the recursive algorithm in python](#)

Bit Mapping

$$S = \{1, 2, 3\}$$

$$P = \begin{matrix} \{\} & \{1\} & \{2\} & \{1, 2\} & \{3\} & \{1, 3\} & \{2, 3\} & \{1, 2, 3\} \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ 000 & 001 & 010 & 011 & 100 & 101 & 110 & 111 \end{matrix}$$

Every possible subset of S can be represented by a binary number with $|S|$ bits.

In the case above, we let 0 represent not in the set and 1 represent an element in the set. We chose the right most digit to correspond with 1, the middle digit to correspond with 2, and the left digit to

correspond with 3. We could have chosen any mapping from the bits to the elements because every possible subset will be represented when you iterate through the binary numbers.

[Click here to view bit mapping algorithm code in python](#)

Comparing Infinities

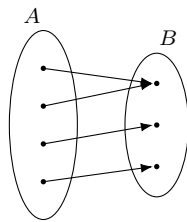
Are some infinities bigger than others?

We need a way to compare the number of elements in two sets.

Consider $A \xrightarrow{\text{mapping}} B$, where A and B are sets.

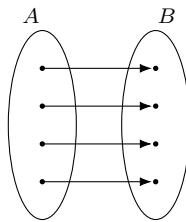
If (all elements in output space are mapped to): $|A| \geq |B|$

Case 1: $|A| > |B|$



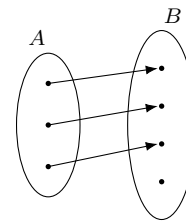
mapping
possible

Case 2: $|A| = |B|$



mapping
possible

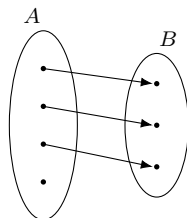
Case 3: $|A| < |B|$



mapping not possible
because each input can
only have 1 output

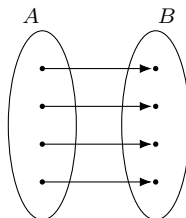
If (different input, then different output):

Case 1: $|A| > |B|$



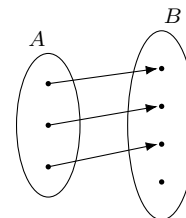
mapping not possible,
leftover input needs dif-
ferent output but no dif-
ferent outputs available

Case 2: $|A| = |B|$



mapping
possible

Case 3: $|A| < |B|$



mapping
possible

If (all elements of output space mapped to
AND
different inputs give different outputs)
THEN $|A| = |B|$ (same number of elements)

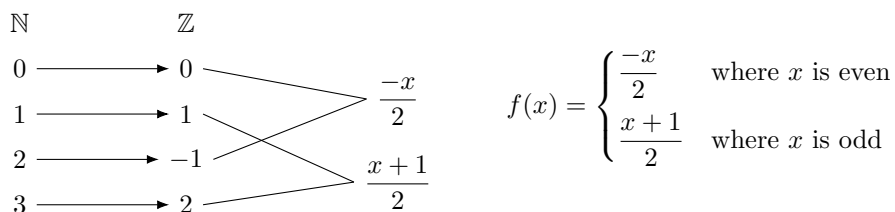
Countably Infinite

A set S is **countably infinite** if there exists a mapping from $\mathbb{N}$ to S that meets the following conditions:

- All elements of output space mapped to

- Different inputs have different outputs

Is $\mathbb{Z}$ countably infinite?



Yes, $\mathbb{Z}$ is countably infinite.

Are the real numbers countably infinite?

Assume $S = \{x \in \mathbb{R} \mid 0 < x < 1\}$ is countably infinite.

Then, there exists a mapping from the natural numbers to S such that all elements in S are mapped to.

0	0. <u>3</u> 4160...	Construct a number such that the number has at least 1 digit different from each real number in list.
1	0.4 <u>8</u> 929...	
2	0.924 <u>5</u> 6...	
3	0.8197 <u>2</u> ...	
4	0.1437 <u>8</u> ...	
⋮	⋮	
		0.56721
		5 ≠ 3 6 ≠ 8 7 ≠ 4 2 ≠ 7 1 ≠ 8

Make sure that the new number does not end with repeating 0's or 9's to avoid, for example, $0.2 = 0.1\bar{9}$.

New number is in S but not in list.

Contradiction, all elements in S should be in list.

S is not countably infinite.

$S \subset \mathbb{R}$ (the set of real numbers is at least as big as S)

$\mathbb{R}$ is not countably infinite.

Russell's Paradox

Are there any restrictions on how you can define a set?

Let $S = \{x \mid x \notin x\}$.

S is the set of all sets that do not contain themselves.

Does S contain itself?

If it does, it must satisfy the condition, $S \notin S$. If it does it doesn't. **Contradiction**

If it doesn't, it satisfies the condition to be in S . If it doesn't it does. **Contradiction**

Since we get a contradiction whether or not S contains itself, we cannot have no restrictions on defining sets.

References

1. Stanford CS103, Lecture 3 slides.
2. Nguyen, W. *COT 3100 Lecture 5 Notes* (handwritten scribe notes), lecture by K. Gajulapalli, 9/8/2026. The typed section of this document, including the questions page, was transcribed directly from those handwritten notes, which appear on the preceding pages of this same PDF.

Questions

1. How can we prove that the bit mapping algorithm can represent all possible subsets?
2. How can we prove the following:

$$(A \setminus B = B \setminus A) \implies A = B$$

3. If there are two different types of stamps, one worth n cents and one worth m cents, what is the minimum value x cents such that any integer cent value of at least x cents can be represented by the stamps?
4. Can you use induction to reverse engineer a solution?
5. Why does rewriting

$$ax^2 + bx + c = 0 \quad \text{as} \quad ax^2 + dx + ex + c = 0$$

where $a, b, c, d, e \neq 0$ and $d + e = b$ and $d \cdot e = a \cdot c$ help you factor?